ARC RESEARCH · THEMATIC RESEARCH NOTE

AI Data Centers Are Going to Space

Google's Project Suncatcher, the TPUs riding SpaceX's Transporter-18, and the five engineering problems between a design paper and an orbital cluster

Power · Cooling · Radiation · Networking · Formation flying · Launch economics

Data as of 24 September 2026 · For educational purposes only, not investment advice

THE THESIS

Orbit swaps AI's power problem for five engineering problems

Google's Project Suncatcher asks whether solar-powered satellites carrying TPUs could one day form a data centre in orbit. Its first test satellite rides a SpaceX rideshare next week. This note walks from why, to how hard, to what is actually flying.

01 · Slides 3-8

Why space

The design paper, AI's power bill, and what changes in orbit

02 · Slides 9-15

Five challenges

Power, cooling, radiation, networking and formation flying, with Google's test data

03 · Slides 16-19

The first flight

The MVP satellite, SpaceX's role, launch cost and the roadmap

04 · Slides 20-22

Who else, and the verdict

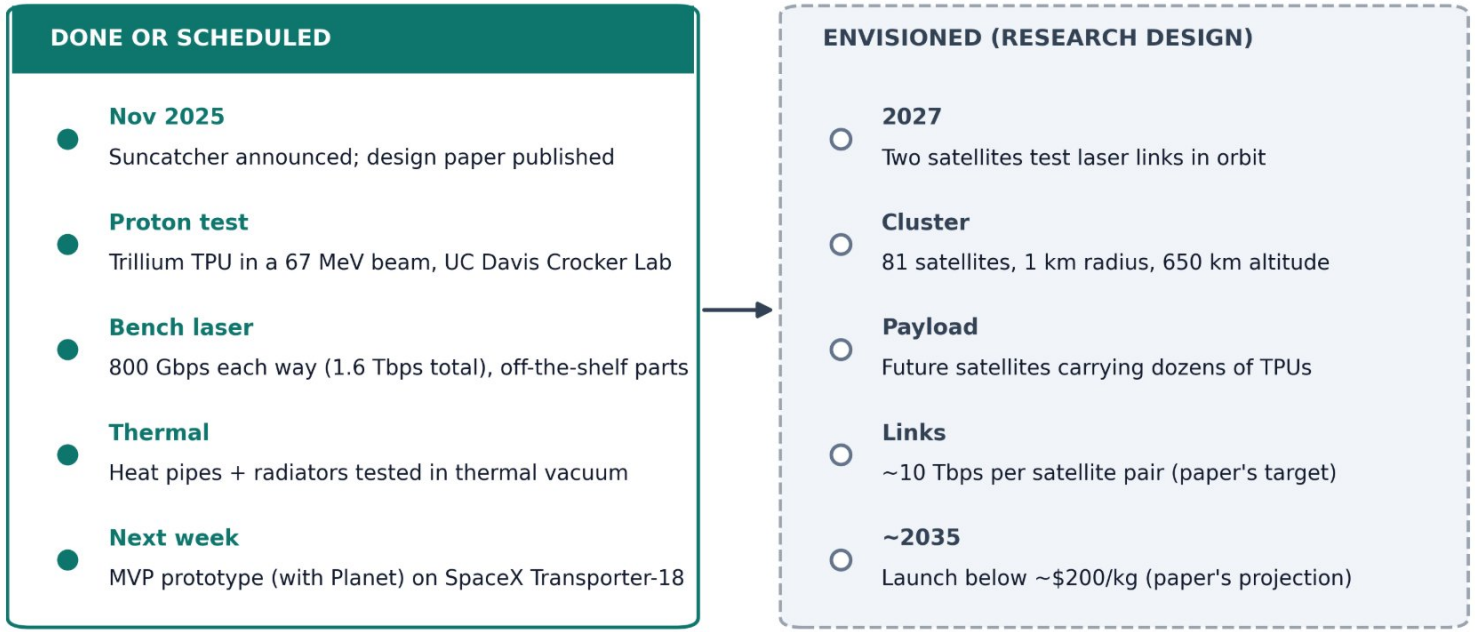
Other orbital-compute efforts and the trade-off that decides it

Suncatcher today: tests and a launch, not a data centre

ARC RESEARCH | PROJECT SUNCATCHER | WHAT IS REAL

Exhibit 1: What Suncatcher has done versus what it only envisions

Everything on the left exists or is scheduled; everything on the right is a research design



Source: Google Research blog (4 Nov 2025); Suncatcher paper, arXiv:2511.19468; blog.google (24 Sep 2026); Ars Technica and NYT via Inc. for the launch date.
Launch date of 1 Oct 2026 is press-reported; Google's post says 'next week'. The ~2035 launch price is the paper's projection, not a quote.

SO WHAT

Google calls Suncatcher a long-term research effort.

The first hardware reaches orbit next week.

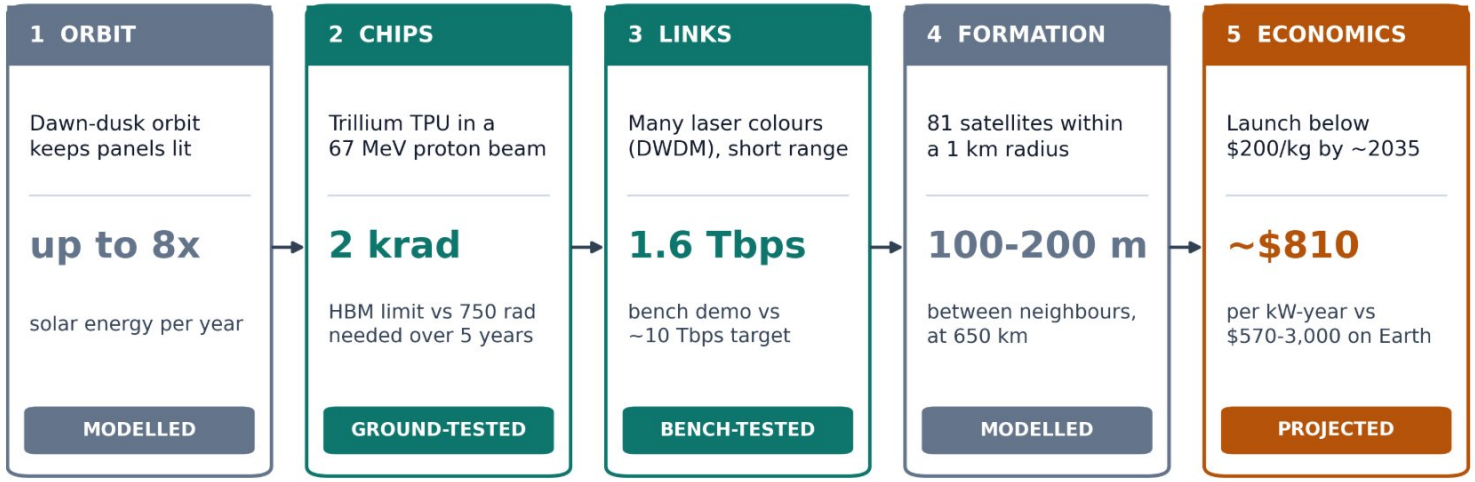
Clusters, 10 Tbps links and cheap launch are still on paper.

How the design paper builds its case

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Exhibit 1A: How the Suncatcher design paper builds its case, step by step

Five linked questions; each answer makes the next one possible, and only two have touched real hardware



Google's reading: nothing in physics or economics rules it out.

Still open: heat rejection at scale, reliability without repair, and laser links between moving satellites.

Source: Suncatcher paper, arXiv:2511.19468 (Nov 2025); Google Research blog (4 Nov 2025). 8x is Google's modelled per-year figure versus a mid-latitude panel. Radiation and laser results are ground and bench tests, not in orbit. ~$810/kW/yr assumes launch below $200/kg (~2035) and is a projection.
Status colours: teal = tested on the ground or bench; slate = modelled; amber = projected. The closing line paraphrases Google's summary of the paper. Schematic.

SO WHAT

Each step unlocks the next: sunlight, then chips, then links, then formation, then cost.

Only the chip and laser steps have been tested, on the ground.

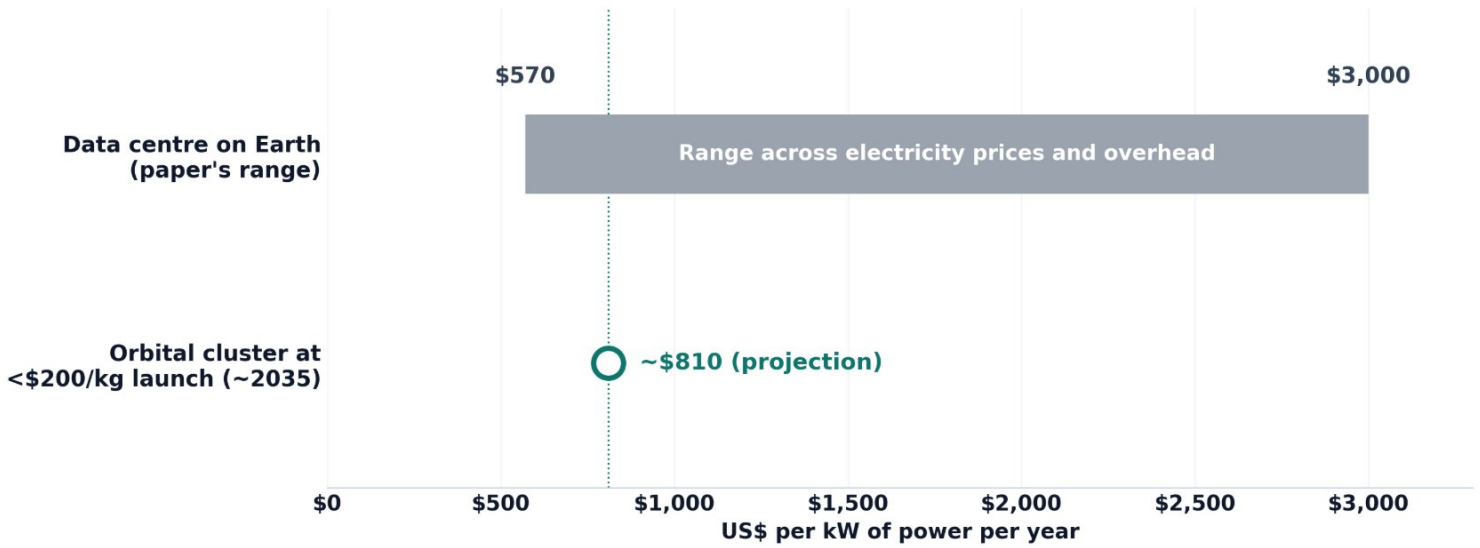
Heat, repair and links in orbit are still open.

The paper's test: competitive with Earth, not cheaper

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Exhibit 1B: The paper's cost test: power cost, US$ per kW per year

If launch falls below $200/kg, the orbital estimate lands inside the range for data centres on Earth, not below it



Source: Suncatcher paper, arXiv:2511.19468: terrestrial data-centre power costs of $570-3,000/kW/y (range reflects electricity prices and facility overhead); orbital estimate of ~$810/kW/y at launch prices below $200/kg, which the paper projects for ~2035 (~20% learning rate, ~180 Starship launches a year). The orbital figure is a projection, not a measured cost.
Bar = range; hollow marker = projection. Linear scale.

SO WHAT

At $200/kg launch, orbital power costs ~$810 per kW-year.

That sits inside the $570-3,000 range on Earth.

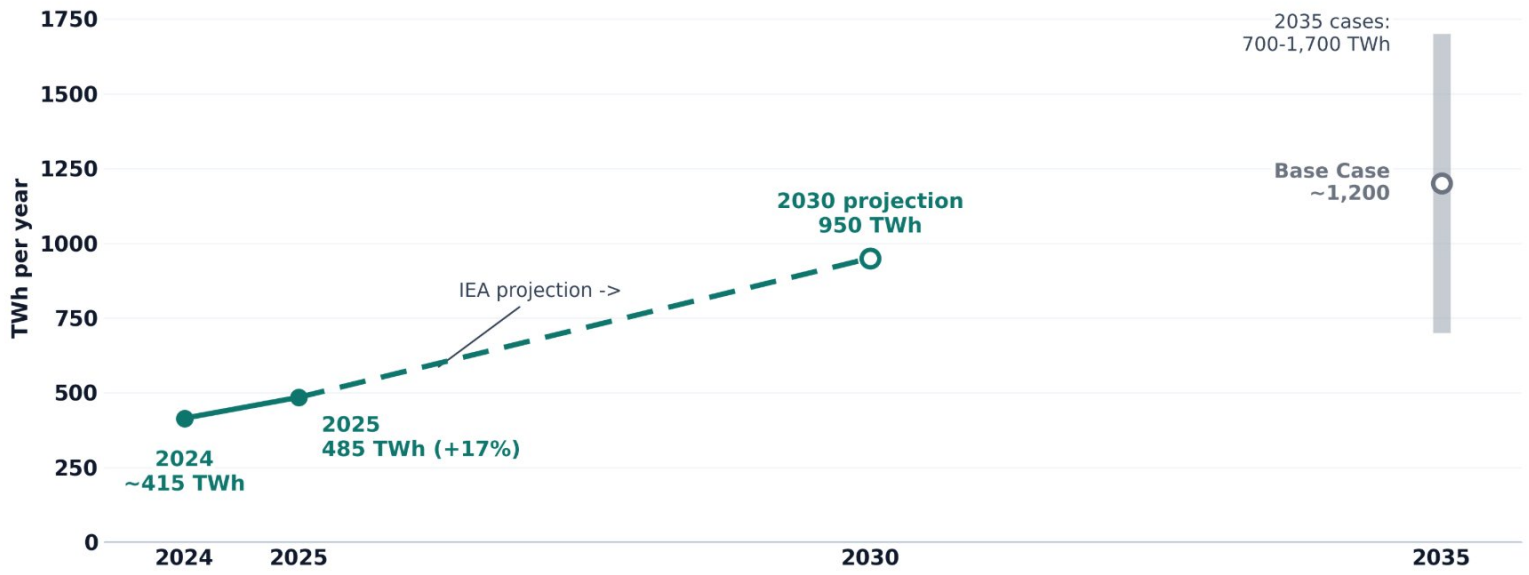
It depends on launch prices falling ~7x by ~2035.

Scaling AI now means scaling power

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Exhibit 2: Global data-centre electricity use, TWh

IEA sees demand roughly doubling between 2025 and 2030, which is why power is the binding constraint



Source: IEA, Energy and AI (Apr 2025) for 2024 (~415 TWh, ~1.5% of world electricity) and the 2035 case range; IEA, Key Questions on Energy and AI (Apr 2026) for 2025 (485 TWh) and 2030 (950 TWh). Two report vintages: the seam checks out (415 -> 485 TWh is +17%, the growth the 2026 report states). The Apr 2025 report put 2030 at ~945 TWh.
Solid = reported estimate; dashed = IEA projection; grey range = IEA's 2035 cases (700-1,700 TWh, Base Case ~1,200 TWh) from the earlier report. IEA values are rounded.

SO WHAT

Data-centre power use rose 17% in 2025, to 485 TWh (IEA).

AI-focused data centres grew about 50% in 2025.

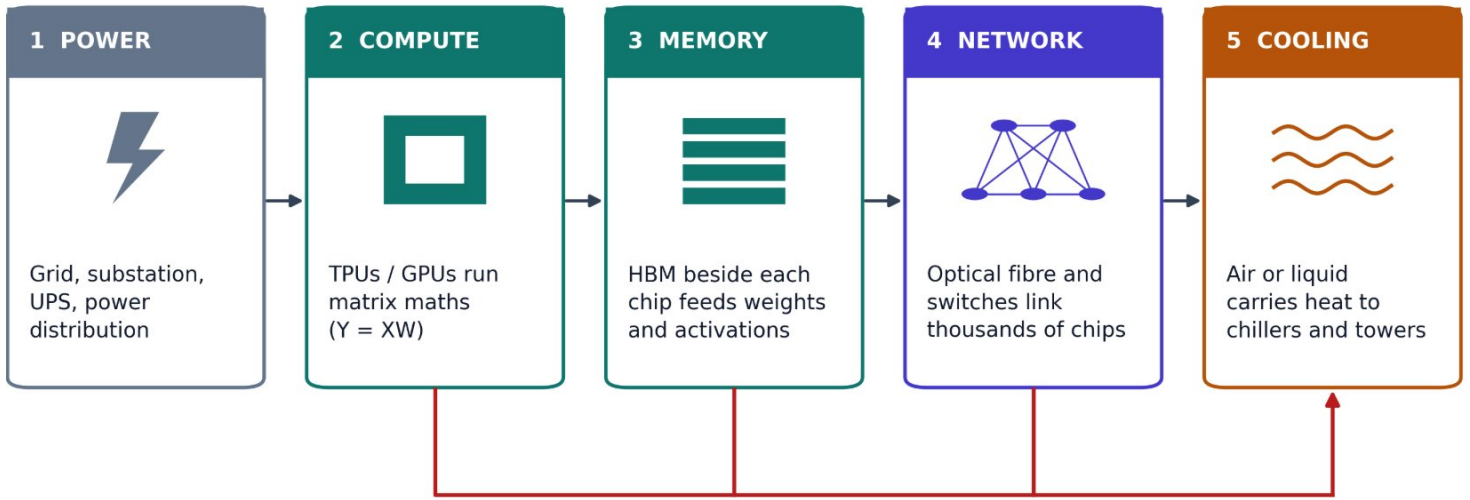
Grid access, not chips, is becoming the constraint Google wants to escape.

A data centre is five systems around the chips

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Exhibit 3: An AI data centre is five systems wrapped around the chips

Almost every watt that powers the accelerators leaves as heat, so power and cooling scale together



On Earth, cooling can use air and water. In orbit there is neither, which is why Exhibit 6 matters.

Schematic, not to scale. HBM = high-bandwidth memory stacked beside each accelerator. Generic description of a terrestrial AI data centre; layouts differ by operator.

SO WHAT

The chips are the easiest part to move.

Power in becomes heat out, so power and cooling scale together.

Networking decides whether thousands of chips act as one computer.

Moving the architecture into orbit

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Exhibit 4: The same five systems, rebuilt for orbit

The chips barely change; everything around them has to be replaced

SYSTEM	ON EARTH TODAY	IN ORBIT (SUNCATCHER CONCEPT)
Power	Grid, substation, UPS	**Solar arrays in a dawn-dusk orbit**
Compute	TPUs / GPUs in racks	**TPUs on each satellite**
Memory	HBM beside each chip	**HBM (radiation-tested)**
Network	Fibre and switches	**Free-space laser links**
Cooling	Air or liquid, chillers	**Heat pipes and radiators**
Structure	Building and racks	**81-satellite formation**

Schematic. Orbit column describes the Suncatcher design concept (arXiv:2511.19468; Google Research blog, 4 Nov 2025), not a built system.
Amber rows are where the physics changes most: no grid, no air, no fibre, and no building to hold it all together.

SO WHAT

Compute and memory carry over, after radiation testing.

Power, network, cooling and structure must be reinvented.

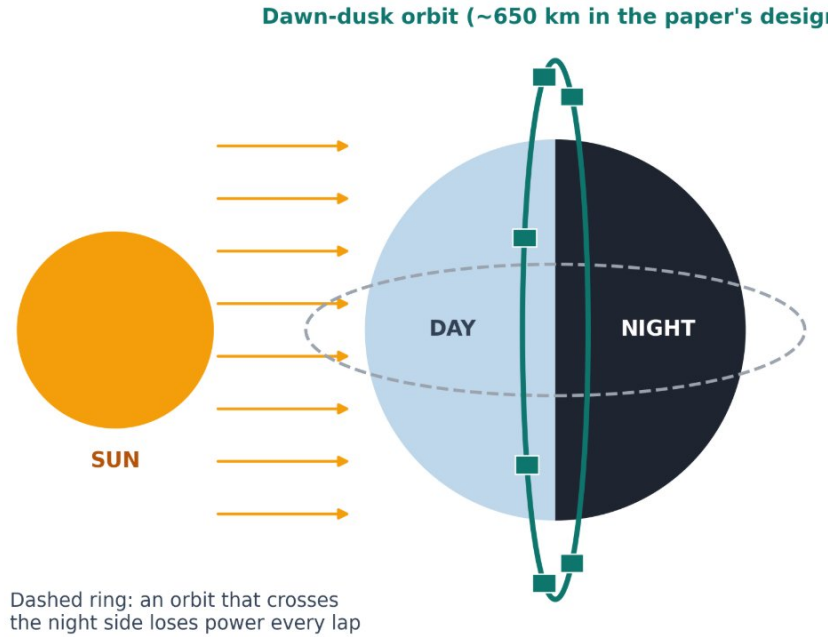
Those four rows are the next five slides.

Near-constant sunlight from a dawn-dusk orbit

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Exhibit 5: A dawn-dusk orbit keeps the panels in near-constant sunlight

Google's paper claims up to 8x more solar energy per year than a panel at mid-latitude on Earth



WHY THIS ORBIT

Orbit plane stays edge-on to the Sun, so satellites ride the day-night line and almost never enter Earth's shadow.

Google claim: up to 8x the energy a mid-latitude panel collects per year.

WHAT A PANEL ON EARTH LOSES

Night: no output between sunset and sunrise
Atmosphere and clouds: less sunlight reaches the cell

Source: Suncatcher paper, arXiv:2511.19468 (dawn-dusk sun-synchronous LEO; 'up to 8x' per year versus a mid-latitude panel on Earth); Google Research blog, 4 Nov 2025.
The 8x figure is Google's own modelled claim, shown as text rather than a bar. Schematic: not to scale; orbit drawn edge-on, so it appears as an ellipse along the terminator.

SO WHAT

Energy is the one thing space makes easier.

Up to 8x more solar energy per year is Google's modelled claim.

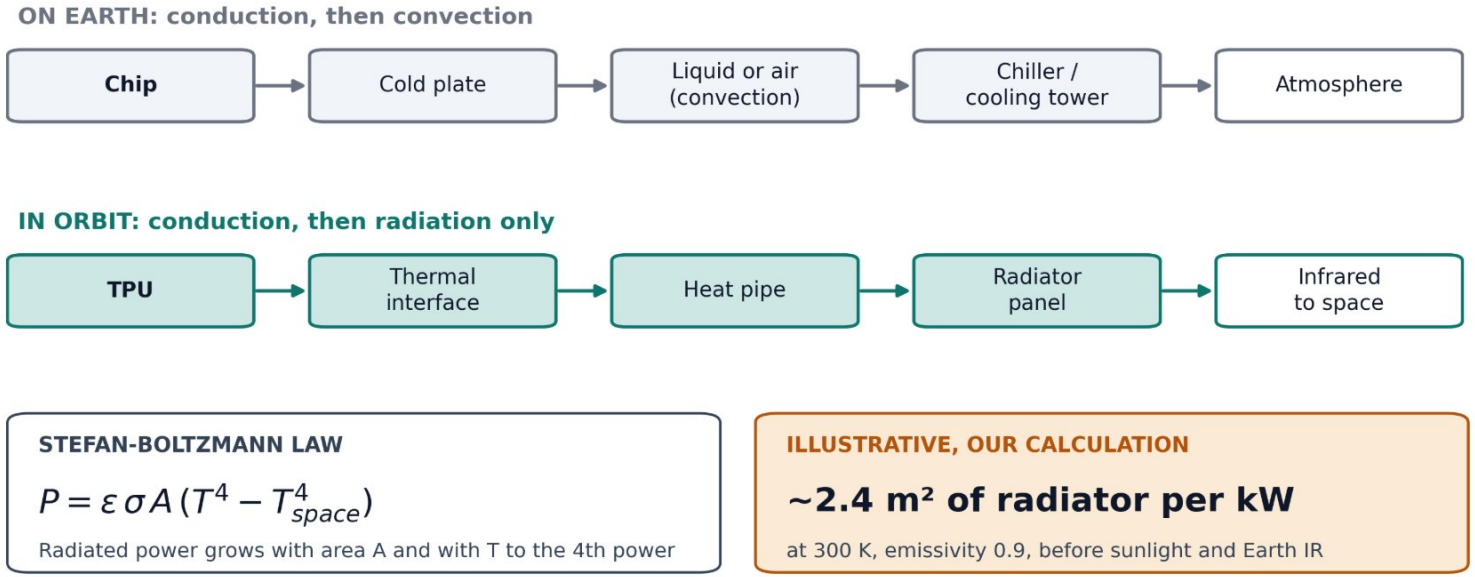
There is no grid to fall back on if the orbit slips.

No air in a vacuum: heat leaves only by radiation

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Exhibit 6: In a vacuum there is no air to carry heat away, only radiators

Every kilowatt of compute needs square metres of radiator, so heat rejection scales with the chips



STEFAN-BOLTZMANN LAW

$$P = \varepsilon \sigma A (T^4 - T_{space}^4)$$

Radiated power grows with area A and with T to the 4th power

ILLUSTRATIVE, OUR CALCULATION

~2.4 m² of radiator per kW

at 300 K, emissivity 0.9, before sunlight and Earth IR

Source: blog.google (24 Sep 2026): heat pipes plus radiators, tested in a thermal-vacuum chamber; Suncatcher paper, arXiv:2511.19468 (thermal system of heat pipes and radiators). Radiator sizing is our illustrative calculation, not a Google figure: Stefan-Boltzmann law, one-sided radiator, emissivity 0.9, 300 K, deep-space sink, ignoring absorbed sunlight and Earth's infrared (both would enlarge it). A = 1,000 W / (0.9 x 5.67e-8 x 300^4) = 2.4 m² per kW. Schematic, not to scale.

SO WHAT

"Space is cold" misleads: a vacuum insulates.

Google uses heat pipes and radiators, tested in a thermal-vacuum chamber.

Radiator area grows with every kilowatt added.

Radiation: single hits now, accumulated dose later

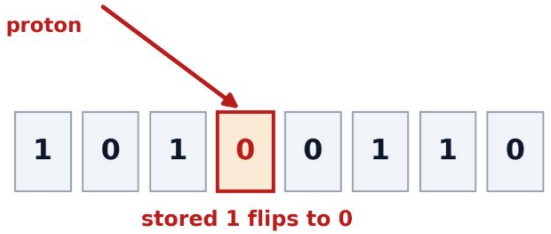
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Exhibit 7: Radiation hurts chips in two ways: single hits and slow accumulation

Single hits flip bits and corrupt answers now; accumulated dose degrades the hardware over years

SINGLE-EVENT EFFECTS (SEE)

One energetic particle, one moment



Bit flip: a wrong value, the chip keeps running

Silent data corruption: a wrong answer nobody sees

Functional interrupt: the chip hangs and needs a reset

Mitigation: error-correcting memory, checks, retries

TOTAL IONIZING DOSE (TID)

Damage that builds up over the mission



Dose adds up every day in orbit

Past the limit, parts drift or fail for good

Mitigation: shielding, tested margin

Source: Suncatcher paper, arXiv:2511.19468 (tested for total ionizing dose and single-event effects with a 67 MeV proton beam); Google Research blog, 4 Nov 2025.
Schematic explainer; particle paths and dose curve are illustrative, not measured data. rad(Si) = radiation dose absorbed by silicon.

SO WHAT

Earth's atmosphere and magnetic field shield chips that orbit does not.

Bit flips can be caught and retried.

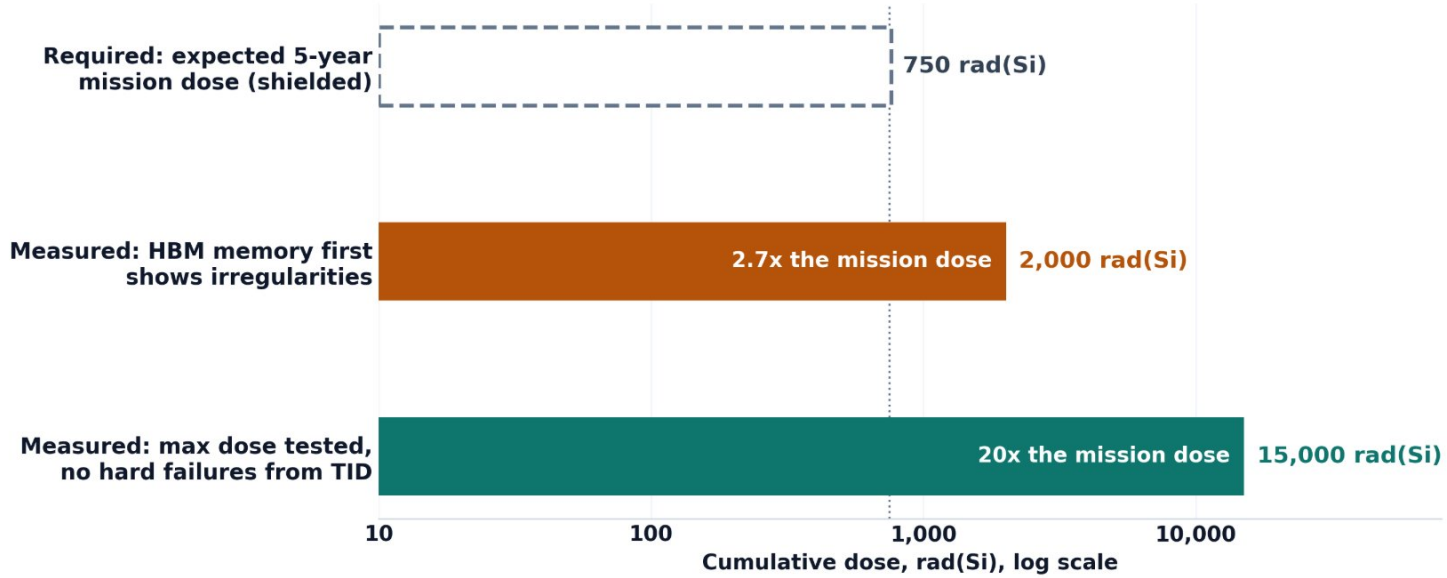
Dose damage is permanent, so margin matters.

Trillium TPUs cleared the 5-year dose with margin

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Exhibit 8: Trillium TPU proton-beam test vs a 5-year mission dose, rad(Si)

The weakest part, HBM memory, held out to ~2.7x the mission dose; nothing failed permanently up to 20x



Source: Suncatcher paper, arXiv:2511.19468: ~750 rad(Si) expected 5-year total ionizing dose (shielded); HBM irregularities after 2 krad(Si); no hard failures attributable to TID up to the maximum tested 15 krad(Si) (one chip). 67 MeV protons, UC Davis Crocker Nuclear Laboratory. Solid = measured in test; outlined = mission requirement (modelled). Multiples are derived (ours): 2,000 / 750 = 2.7x, 15,000 / 750 = 20x. Log scale.

SO WHAT

HBM memory is the weak point, at 2.7x the mission dose.

The 15 krad result is from a single chip.

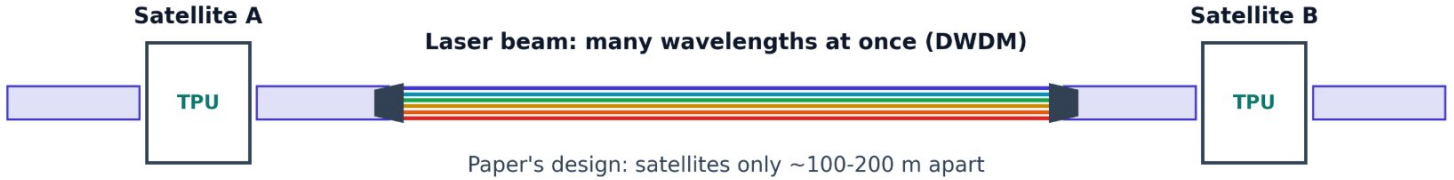
A proton beam is not orbit: MVP gets the real test.

Lasers replace fibre between satellites

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Exhibit 9: Lasers replace fibre: data is carried on light across open space

Many wavelengths share one beam (DWDM), and satellites fly close so enough light arrives



ENCODE	MULTIPLY	AIM
Data is put on light by an optical transceiver, as in fibre, but the light crosses open space.	Up to 24 wavelengths per aperture (DWDM) plus several apertures (spatial multiplexing) add up.	Both ends move at orbital speed; the beam must stay locked on a coin-sized target from miles away.

Source: Suncatcher paper, arXiv:2511.19468 (free-space optical links; 24 DWDM channels, half the C-band, per aperture; satellites hundreds of metres apart); Quartz and Interesting Engineering, 24 Sep 2026 (Google's own comparison: hitting a coin-sized target from miles away while both ends move). Schematic, not to scale.

SO WHAT

Training splits a model across chips that constantly swap data.

A slow link leaves expensive chips waiting.

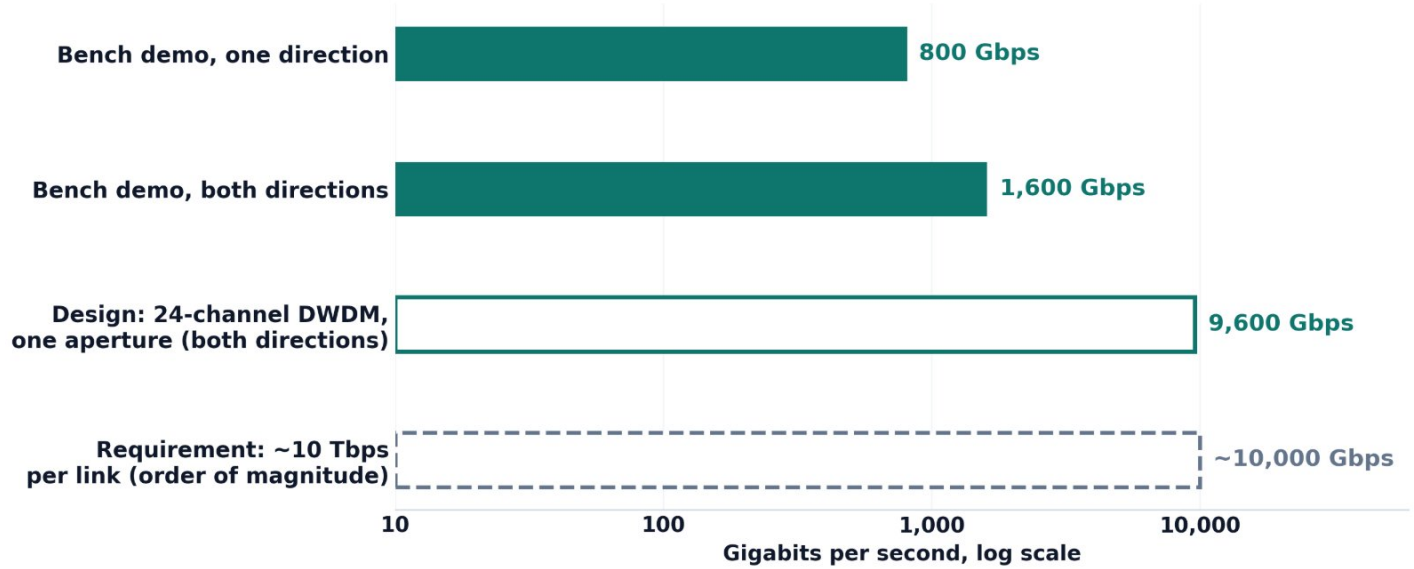
In orbit the network is part of the computer.

The bandwidth gap: about 6x, and not yet in orbit

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Exhibit 10: Inter-satellite link bandwidth, demonstrated versus needed, Gbps

The bench demo reached 1.6 Tbps; the design needs roughly 10 Tbps per link, about 6x more, in orbit



Source: Suncatcher paper, arXiv:2511.19468: bench demonstrator with off-the-shelf parts, 800 Gbps one way (1.6 Tbps bidirectional); 24 DWDM channels per aperture yield 9.6 Tbps bidirectional (design calculation); required aggregate bandwidth per link 'on the order of 10 Tbps'. Google's public posts describe the need as tens of Tbps. Solid = demonstrated on a bench; hollow = paper's design calculation; dashed outline = paper's stated requirement (order of magnitude). 10,000 / 1,600 = 6.25x is derived (ours). Log scale.

SO WHAT

1.6 Tbps was shown on a lab bench with off-the-shelf parts.

The paper's design needs ~10 Tbps per link.

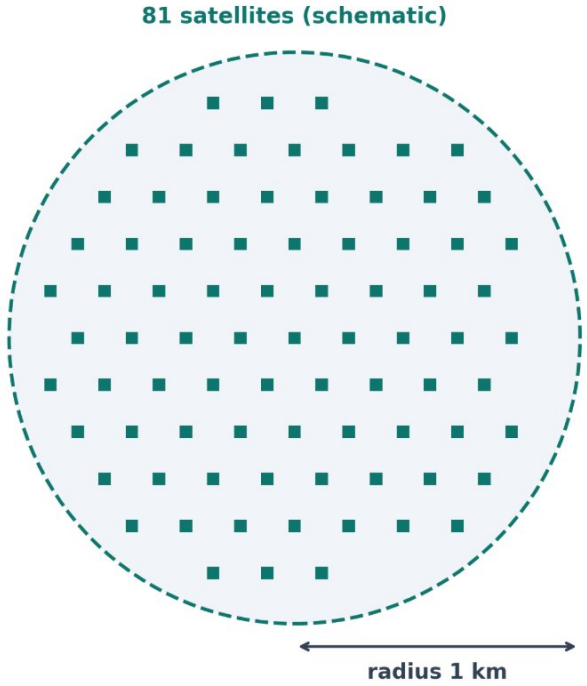
Links between moving satellites wait for the 2027 test.

81 satellites flying hundreds of metres apart

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Exhibit 11: The paper's illustrative cluster: 81 satellites inside a 1 km radius

Neighbours sit only ~100-200 m apart, close enough for fat laser links, close enough to make collisions a design issue



81

satellites in the illustrative cluster

1 km

cluster radius

~100-200 m

between neighbours, oscillating every orbit

650 km

mean altitude, dawn-dusk orbit

WHY SO TIGHT

Received laser power falls off fast with distance; flying close keeps 10 Tbps-class links feasible, but every satellite must hold station without colliding.

Source: Suncatcher paper, arXiv:2511.19468: illustrative planar 81-satellite constellation, mean altitude 650 km, cluster radius R = 1 km, next-nearest-neighbour distance oscillating between about 100 and 200 m over each orbit. Dot positions are our schematic packing of 81 points in a 1 km-radius disc, not the paper's modelled positions.

SO WHAT

Close spacing is what makes 10 Tbps links plausible.

It also turns the cluster into a collision-avoidance problem.

Supercomputer design meets orbital mechanics.

Meet MVP, the test satellite flying next week

Item	Detail	Who says so
Chips	Google Trillium TPUs	Google
Number of TPUs	4	Press (NYT via Inc.)
Spacecraft	Prototype developed with Planet	Google
Ride	SpaceX Transporter-18 rideshare	Google
Launch	~1 Oct 2026, Falcon 9, Vandenberg	Press; Google: "next week"
Power	~1 kW from solar panels	Press (NYT via Inc.)
Compute	About one data-centre server	Press (NYT via Inc.)
Launch loads	Up to 10 g; 50-100 g on chips	Google

WHAT IT WILL MEASURE

Whether TPUs survive launch shaking and acceleration.

How they run under real radiation and heat swings.

Its goal is engineering data, not a working data centre.

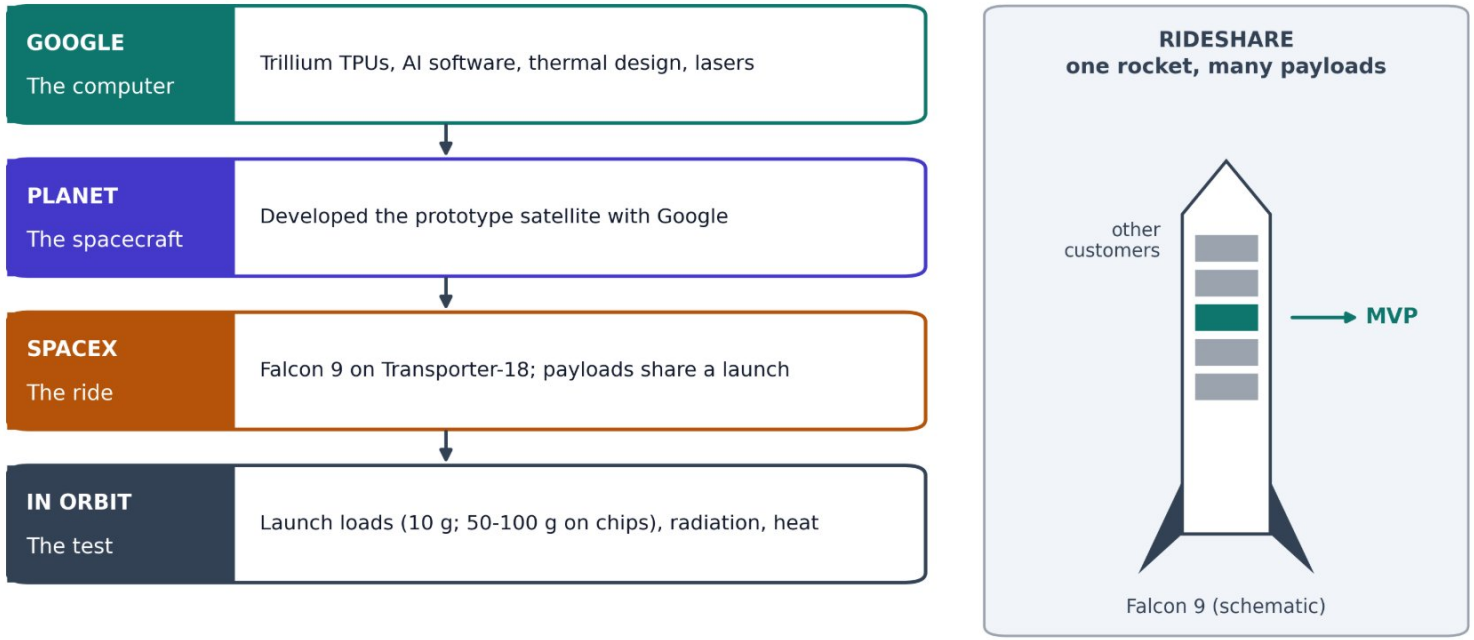
Google's 24 Sep post does not state the TPU count, date, site or power; those rows are press-reported and labelled so.

SpaceX is the transportation layer

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Exhibit 12: Who does what on the first flight: Google, Planet, SpaceX

SpaceX is not building the orbital data centre; for MVP it is the ride, on a shared rocket



Source: blog.google, 24 Sep 2026 (Transporter-18 rideshare with SpaceX, developed with Planet; up to 10 g sustained, 50-100 g on components such as TPU chips; heat pipes and radiators; Trillium TPUs). Falcon 9 from Vandenberg and the 1 Oct date are press-reported (Converge Digest, Ars Technica via RuntimeWire). Schematic, not to scale.

SO WHAT

Near term: a rideshare seat means no dedicated rocket for a test.

Long term: every panel, chip and radiator must be launched.

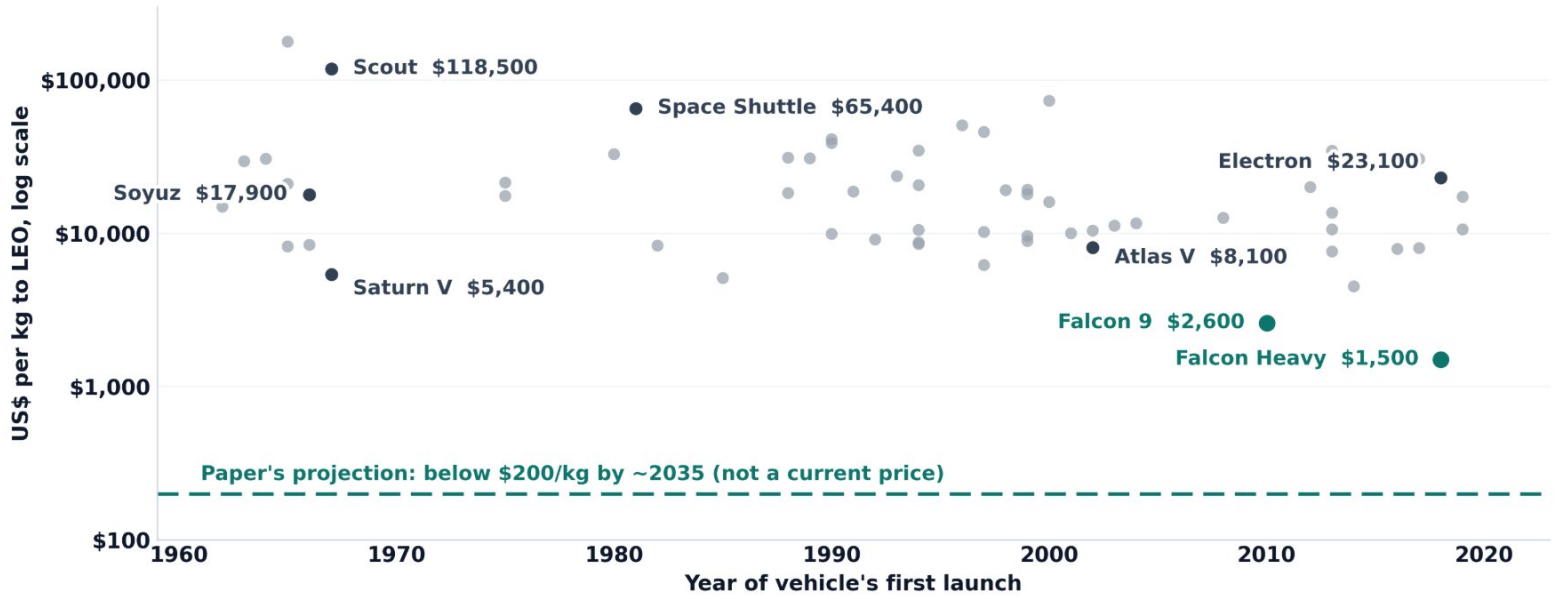
So the price of a kilogram to orbit sets the ceiling.

Launch cost has to fall another ~7x

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Exhibit 13: Cost to launch to low Earth orbit by vehicle, US$ per kg (2021 dollars)

SpaceX's Falcon rockets set the low end of the range; Google's case needs a further ~7x drop, to below $200/kg



Source: CSIS Aerospace Security Project (2022), via Our World in Data (constant 2021 US$ per kg of payload to LEO, plotted at each vehicle's first-launch year; 61 vehicles). $200/kg by ~2035 is the Suncatcher paper's projection (arXiv:2511.19468; assumes a ~20% learning rate and ~180 Starship launches a year), not a price anyone charges today. Starship has no CSIS figure and is not plotted. 1,500 / 200 = 7.5x is derived (ours). Log scale.

SO WHAT

Falcon Heavy: $1,500/kg in 2021 dollars (CSIS).

At $200/kg the paper models ~$810 per kW-year in orbit, versus $570-3,000 on Earth.

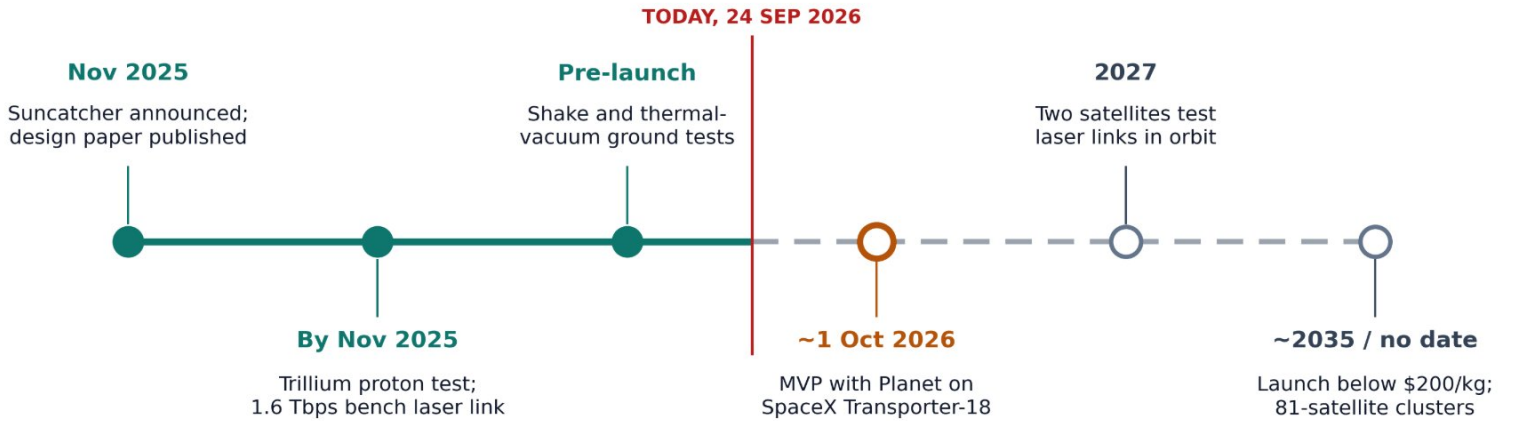
That needs ~180 Starship launches a year: a projection.

From design paper to cluster

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Exhibit 14: From design paper to cluster: Suncatcher milestones

Everything left of today is done; the first in-orbit test is days away; clusters remain a concept with no date



Source: Google Research blog (4 Nov 2025); Suncatcher paper, arXiv:2511.19468; blog.google (24 Sep 2026: shake, thermal-vacuum and radiation tests; 2027 two-satellite test).
1 Oct 2026 launch date is press-reported (Ars Technica, NYT via Inc.). ~2035 is the paper's launch-price projection. Filled = done; hollow = scheduled or planned; dashed = projection or concept.
Spacing is not to scale in time.

SO WHAT

MVP answers one narrow question: do TPUs work in orbit?

2027 tests the hardest piece: laser links between satellites.

Clusters have no public date.

WHO ELSE · ORBITAL COMPUTE EFFORTS

Google is not alone: three others already have hardware up

Who	What has flown	What is planned	Status
Starcloud	Starcloud-1, Nov 2025: one Nvidia H100	Commercial access via Crusoe, 2027	Flown
ADA Space / Zhejiang Lab	12 computing satellites, May 2025	Up to 2,800 satellites	Flown
Axiom Space + Spacebilt	Two data-centre nodes, Jan 2026	More nodes on Kepler's network	Flown
SpaceX (with xAI)	None as data centres	FCC filing (Jan 2026): up to 1M satellites	Filing
Blue Origin	None	FCC filing for ~51,600 satellites	Reported
Cowboy Space (ex-Aetherflux)	None	First node targeted early 2027; $275M raised	Planned

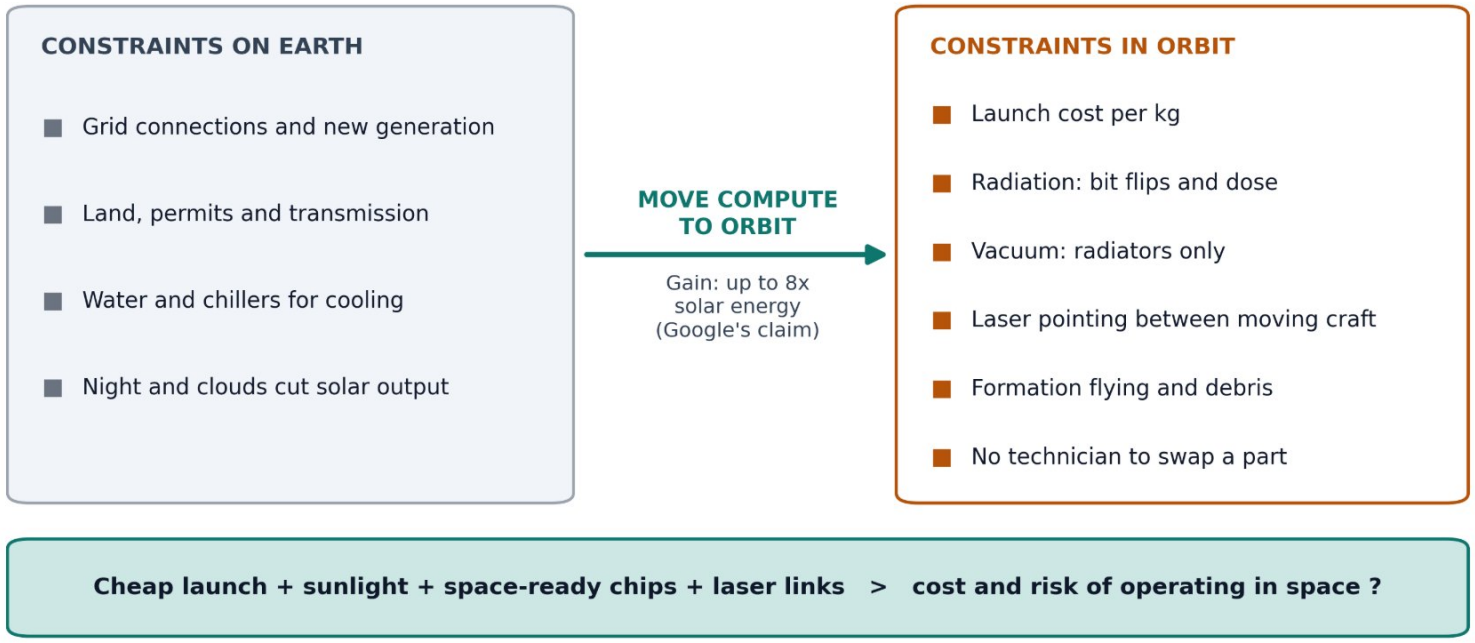
Flown = hardware in orbit per company or state sources. Filing = regulatory application, not a deployment plan. Reported = press reading of a filing. Satellite counts in filings are licensing ceilings. Sources on slide 22.

When does sunlight outweigh everything space makes harder?

ARC RESEARCH | PROJECT SUNCATCHER | THE BIG QUESTION

Exhibit 15: Suncatcher swaps Earth's constraints for space's constraints

The question is not whether space is better, but when abundant sunlight outweighs everything that gets harder



Source: ARC Research framing of the trade-offs described in the Suncatcher paper (arXiv:2511.19468), Google Research blog (4 Nov 2025) and blog.google (24 Sep 2026).
LEO debris count: ~44,870 tracked objects, as reported by Quartz (24 Sep 2026). Qualitative comparison; no weighting implied by list length.

SIGNPOSTS TO WATCH

MVP flight data on TPU errors, heat and uptime.

Whether the 2027 laser links hold between moving satellites.

Starship cadence and the price per kg.

SOURCES

Where every figure comes from

GOOGLE (PRIMARY)

Suncatcher paper, "Towards a future space-based, highly scalable AI infrastructure system design", arXiv:2511.19468 (Nov 2025)

Google Research blog, "Exploring a space-based, scalable AI infrastructure system design" (4 Nov 2025)

blog.google, "Learn about Google's Project Suncatcher to put ML infrastructure in space" (24 Sep 2026)

DATA

IEA, Energy and AI (Apr 2025); IEA, Key Questions on Energy and AI (Apr 2026)

CSIS Aerospace Security Project (2022), launch cost to LEO, via Our World in Data

PRESS (LABELLED WHERE USED)

Inc. citing The New York Times; Converge Digest; RuntimeWire citing Ars Technica; Quartz; Interesting Engineering (all 24 Sep 2026)

OTHER EFFORTS (SLIDE 18)

Starcloud and Nvidia posts; CNBC (Dec 2025); SpaceNews (May 2025); Axiom Space release (Jan 2026); FCC filing DA-26-113; Data Centre Magazine (Mar 2026); DCD (May 2026)

OUR CALCULATIONS

Radiator area, dose multiples, the ~6x bandwidth and ~7x launch-cost gaps: formulas in each exhibit's footer